

Building Your First AI Agent

Agents, Prompts, Tools, Context & Escalations — explained for builders



What We'll Cover Today

01

What is an Agent?

From workflows to reasoning systems

02

Types of UiPath Agents

Autonomous, Conversational, Coded

03

Anatomy of an Agent

The building blocks inside Agent Builder

04

System & User Prompts

Telling the agent who it is and what to do

05

Tools & Context

How agents act and stay grounded in facts

06

Escalations & Best Practices

Human-in-the-loop and a worked example

Explorer

- Solution 5 ✓
- Agent
 - Definition
 - Evaluation Sets
 - Evaluators

Select agent type

LOW-CODE
Build visually in Studio

Autonomous
Runs a task end to end on its own.

Conversational
Chats with users in real time.

CODE
Build in your IDE · Start with sample

Coded Preview
Run **autonomous** or **conversational** coded agents

Get started with Autopilot

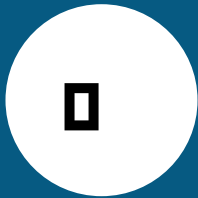
What should your agent do?

Describe the agent you want to create...

Start fresh Generate Agent

- Apps
- Assets
- Connections
- Indexes
- IXP Models
- Processes
- Queues
- Storage buckets
- Task catalogs

Types of UiPath Agents



Autonomous

Triggered by time or a system event; runs a task end-to-end from a single prompt.

e.g. reading an inbox, extracting invoice data, updating an ERP



Conversational

Handles real-time, multi-turn dialogue — tracking context across the whole chat.

e.g. an HR assistant answering follow-up policy questions



Coded

Built in Python (LangChain / LangGraph / LlamaIndex) for full developer control.

e.g. a custom triage agent orchestrated inside Maestro

UiPath Studio

Solution 5 > Agent

Saved

Build

Manage

Debug

Debug step-by-step

Publish

Submit as template

Share

Explorer

Solution 5

Agent

Definition

Evaluation Sets

Evaluators

Score: NA

0%

Clone as Coded Agent

Preview

Canvas

Form

Autonomous

General

Arguments

System prompt * ⓘ

Edit Preview H B I <> | Coverage |

You are an agentic assistant.

User prompt * ⓘ

Edit Preview H B I <> | Coverage |

What is the current date?

Memory space Preview

Add memory space

Add memory spaces the agent can use.

Tools

Add tool

Enable the agent to access data and perform actions.

Contexts

Add context

Add contexts the agent can query.

Escalations

Add escalation

Escalate to an assigned person before the agent takes an important action.

Not sure how to start? Take a tour

Agent (Autonomous Agent)

Standard harness

Advanced harness Preview ⓘ

Model *

gpt-5.4 Suggested

Note: Selecting certain AI models may require a confirmation step.

Model settings

Temperature

Precise Creative

Max. tokens per response

0 128000

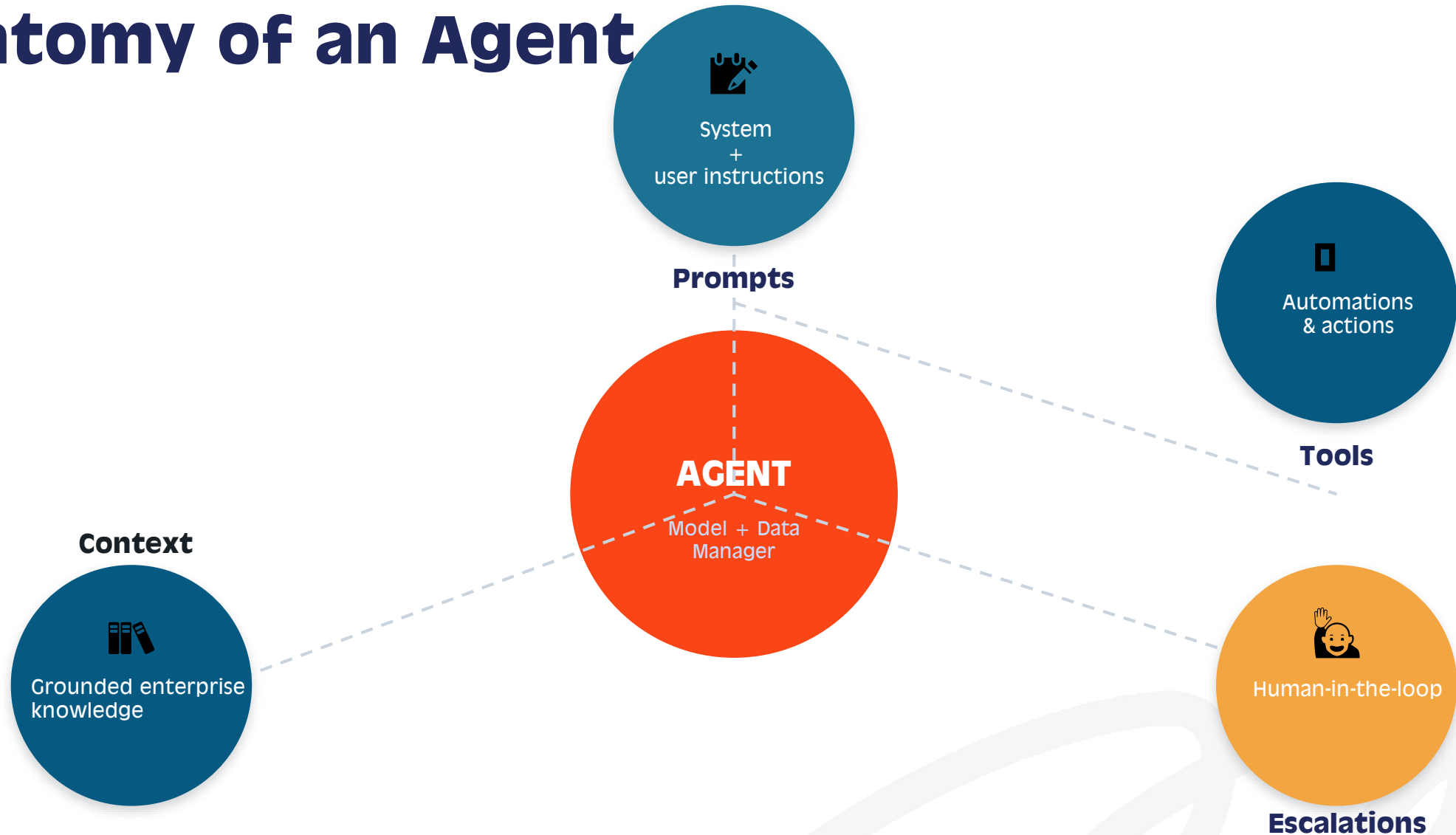
Max. iterations

1 100

Guardrails ⓘ

Add guardrails

Anatomy of an Agent



Studio Web lets you author the same agent in Form view (structured panels) or Canvas view (drag-and-drop nodes) — kept in sync.

The System Prompt



Sets the agent's identity, goal, and rules of engagement — in plain natural language.

- Defines role, goal, and constraints for the agent
- States when it may use certain tools, context, or escalations
- Forms the plan the agent follows and adapts over time
- Should give a specific persona, not a generic assistant

EXAMPLE — SYSTEM PROMPT

“You are a refund processing agent for abc Retail. You only help customers with refund requests and politely decline anything else.

1. Read the customer email.
2. Identify the order ID.
3. If found, use the Find Order Details tool to check eligibility.
4. If approved, draft a confirmation message.
5. If unsure, raise a human escalation before responding.”

The User Prompt & Arguments



Carries the per-run task: the actual case the agent must act on right now.

- Arguments are defined in the Data Manager (Arguments tab)
- Referenced inside the user prompt with `{{argumentName}}` syntax
- Lets the agent take structured inputs, like an activity does
- System prompt = constant rules | User prompt = this run's task

EXAMPLE — USER PROMPT

“Process this refund request:

Customer email: {{customerEmail}}

Order ID: {{orderId}}

Reason stated: {{refundReason}}

Follow the steps in your instructions and respond with the outcome.”

Arguments make the agent reusable — the same instructions run for every customer, order, and reason.

Tools

Tools are how an agent's reasoning turns into action. Without a tool, an agent can only reason and talk — not act.



Automations

Existing UiPath processes and workflows the agent can invoke as a step



Integration Service

Prebuilt connectors and activities for enterprise apps and APIs



Other Agents

One agent calling another — the basis of multi-agent systems

Best practice: give every tool a focused name and description of its purpose — that's how the agent learns when to call it, from the system or user prompt.

Context (Context Grounding)



Connects the agent to enterprise knowledge — so answers are grounded in real documents, not guesses.

- Links Context Grounding indexes (RAG over your data sources)
- You configure the query strategy, thresholds, and retrieval limits
- The agent retrieves only what's relevant to the current step
- Least-context principle: pass only essential context to the LLM



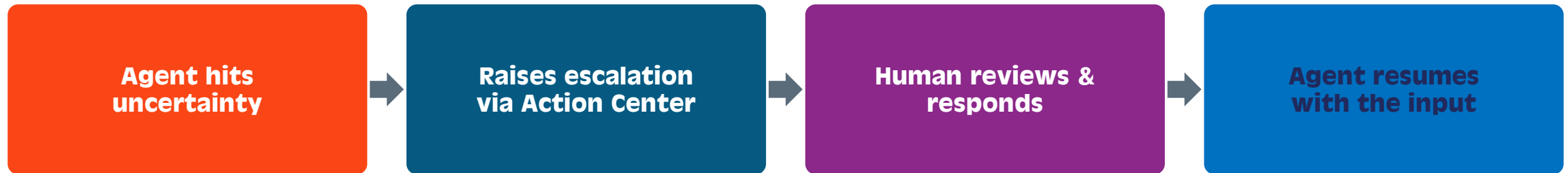
WHY IT MATTERS

Reduces hallucination — answers cite real enterprise data.

Keeps the agent fast and cheap — only relevant chunks are retrieved, not entire documents.

Escalations

Escalations let an agent pause and hand a decision to a human when it can't confidently proceed on its own.



Built with Action Apps

Escalations run through an Action Center app that presents the case and captures the decision.

Outcome behavior

Continue resumes the agent with the human's input; End terminates the run right there.

Referenced in the prompt

e.g. "Before using Tool X, raise the Human Confirmation escalation."

Agent Memory (optional)

Remembers past human answers (up to a 3-month TTL) so the same question isn't asked twice.

RECAP

Five Building Blocks of an Agent



System Prompt

Identity & rules



User Prompt

This run's task



Tools

Turns thought into action



Context

Grounds it in facts



Escalations

Brings in a human

Try it yourself: Studio Web → Agents → New Agent (Form or Canvas view)

Docs: docs.uipath.com/agents

Any Queries? 🙋

Thank you! — See you again in Chapter 3!

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